# José Manuel Requena Plens

Firmware & Software Engineer

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On-site, remote or hybrid

## Summary

**Embedded firmware and software engineer** with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid R&D background in acoustics (physics). Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

## Experience

**R&D Software Developer** — **Apr 2023 - May 2026**
*Power Electronics S.L. — Solar + Storage* — *Lliria, Spain*

- Reduced RTOS task stack usage by 30%–50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ~50–80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library mocks/fakes, plus hundreds of on-hardware integration tests (pytest) — TCP/FTP performance and FTP security —; all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

**Predoctoral researcher** — **Jul 2021 - Jul 2022**
*UPV · CSIC · i3M* — *Valencia, Spain*

- Designed acoustic metamaterials (metadiffusers and lenses) that reduce the footprint of conventional diffusers, validated with MATLAB and COMSOL simulations.
- Manufactured and characterized prototypes; published results at international conferences (Euronoise, Tecniacústica).
- Supervised a Master's thesis in Acoustic Engineering (UPV), from research design to defense.

**Researcher in European project** — **Feb 2020 - Jul 2021**
*UPV · European Space Agency (ESA) — IGIC* — *Gandia, Spain*

- Designed and experimentally validated a noise-mitigation metamaterial for the VEGA rocket launch (ESA project AO/1-9479/18/NL/LvH), meeting 100% of project objectives with results accepted by ESA.
- Developed acoustic measurement software per ISO 10534-2 and ASTM 2611 (A|Lab) and a 3-axis robotic system for experimental measurements.
- Co-author of international publications/talks (Euronoise, ECSSMET).

## Technical Skills

**Embedded Firmware & Electronics:** C, STM32 & ARM, ESP32 / ESP-IDF, PlatformIO, Texas Instruments, FreeRTOS / RTOS, Comm. Protocols, HW Debugging, Keil uVision, Instrumentation, Secure Boot & Embedded Security, Doxygen

**Software Engineering & Tools:** Python, C++, Go, TypeScript, Bash, Clang Tooling, Git, GitHub, GitLab, Docker, MATLAB, CMake, JetBrains Toolbox

**Quality, CI/CD & DevSecOps:** GitHub Actions, GitLab CI, Jenkins, SonarQube / SonarCloud, GoReleaser, Static Analysis & SAST, Testing, pre-commit hooks

**Networking:** TCP/IP & Networking, MikroTik RouterOS, WireGuard / VPN, IPv6 & Dual-stack, QUIC / HTTP3, DNS / Dynamic DNS

**Security & Cryptography:** Applied Cryptography, TLS / mTLS / PKI, Nginx Hardening, CrowdSec, Network Hardening

**AI & Agentic Engineering:** Model Context Protocol (MCP), AI Coding Agents, LLM Integration

**Simulation & Acoustics:** COMSOL Multiphysics, LabView, Acoustic Instrumentation

**General & Platforms:** Linux / Unix, MacOS, Windows, LaTeX, HTML / CSS

## Open Source Projects

**gitlab-mcp-server** — Go · Model Context Protocol server
45 stars • 58 releases • ~158k downloads • On npm, PyPI & NuGet; listed on Glama, mcp.so, Cursor & mcpservers.org
Open-source GitLab MCP server that exposes 850+ GitLab actions (1,000+ depending on edition and deployment) to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

**phonometry** — Python · DSP · Acoustics
128 stars • 28 releases • PyPI
Standards-conformant acoustics library: sound level metrology, psychoacoustics, room, building, environmental and underwater acoustics, vibration, electroacoustics and wave simulation — every method implemented from the standard, with 1893 CI-enforced conformance checks across 111 domains and 523 standards. Formerly PyOctaveBand.

**cs-routeros-bouncer** — Go · Firewall · MikroTik · WAF
21 stars • 21 releases • On the CrowdSec Hub
CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

**Cloudflare-DNS-Updater** — Bash · DevOps
29 stars • 7 releases
Dynamic DNS client for Cloudflare (IPv4/IPv6) written in Bash, with a bats test suite and coverage reporting.

## Education

**PhD program in Health and Wellness Technologies (discontinued)** — **2019–2023**
*Polytechnic University of Valencia (i3M / UMIL)* — *Valencia, Spain*
Research on acoustic metamaterials for medical imaging and ultrasound beam control (programme not completed; redirected to industrial software development).

**Master's Degree in Acoustical Engineering** — **2019**
*Polytechnic University of Valencia (EPSG)* — *Gandia, Spain*
Graduated with honors; Master's thesis awarded a special distinction.

**Degree in Telecommunication Engineering in Sound and Image** — **2018**
*University of Alicante (EPS)* — *Alicante, Spain*
Two specializations: Acoustic Engineering and Audiovisual Technology.

**Higher Vocational Degree — Superior Sound Technician** — **2011**
*IES Luis García Berlanga* — *Alicante, Spain*

**Intermediate Vocational Degree — Electronics Technician** **2009**
*IES Salesianos (San José Artesano)* *Elche, Spain*